IGCSE 0607 Practice Questions — E1.1 — Combined Questions & Markscheme

Types of number

Total Questions: 21 — **Total Marks:** 130 Marks

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- **E1.1** Types of number
  → **Paper 2 (Non-Calculator)** *(14 Qs)* · **Paper 4 (Calculator)** *(7 Qs)*

1 **(a)** Express 504 as a product of its prime factors.

..... **[2]**

(b) Find the smallest positive integer k such that $504k$ is a square number.

$k =$ ..... **[2]**

2 **(a)** Write 84 as a product of its prime factors.

..... **[2]**

(b) Find the lowest common multiple (LCM) of 84 and 126.

..... **[2]**

3 Here is a list of numbers.

2 9 12 23 36 45 64 81

(a) Write down a number from the list which is:

(i) a multiple of 12,

..... **[1]**

(ii) a cube number,

..... **[1]**

(iii) both a prime number and a factor of 46.

..... **[1]**

(b) Find the lowest common multiple (LCM) of 36 and 64.

..... **[2]**

4 Here is a list of numbers.

$$0.25 \quad \sqrt{7} \quad 8 \quad 9 \quad 13 \quad -5 \quad 21 \quad 64$$

(a) From the list, write down a prime number.

..... **[1]**

(b) From the list, write down an irrational number.

..... **[1]**

(c) From the list, write down an integer less than zero.

..... **[1]**

(d) From the list, write down the reciprocal of 4.

..... **[1]**

(e) From the list, write down a number that is both a square number and a cube number.

..... **[1]**

5 Here is a list of numbers.

8 15 23 27 36 49 64

(a) From the list, write down

(i) a prime number,

..... **[1]**

(ii) a cube number,

..... **[1]**

(iii) a number that is both a square number and a cube number.

..... **[1]**

(b) Express 36 as a product of its prime factors.

..... **[2]**

6 Here is a list of numbers.

2 9 15 16 23 27 64 72

(a) From the list, write down

(i) a prime number,

..... **[1]**

(ii) a cube number,

..... **[1]**

(iii) a number that is both a square number and a cube number,

..... **[1]**

(iv) a triangle number.

..... **[1]**

(b) Express 72 as a product of its prime factors.

..... **[2]**

7 **(a)** Write 168 as a product of its prime factors.

..... **[2]**

(b) Find the highest common factor (HCF) of 168 and 252.

..... **[2]**

(c) Find the lowest common multiple (LCM) of 168 and 252.

..... **[2]**

8 Here is a list of numbers.

$$-5 \qquad -1 \qquad \frac{3}{8} \qquad 8 \qquad 13 \qquad 25 \qquad \sqrt{30} \qquad \pi$$

(a) From this list, write down:

(i) a prime number,

..... **[1]**

(ii) a cube number,

..... **[1]**

(iii) an irrational number,

..... **[1]**

(iv) an integer that is not a natural number.

..... **[1]**

(b) Find the reciprocal of 0.8.
Give your answer as a fraction in its simplest form.

..... **[2]**

9 Two numbers, A and B, are written as products of their prime factors.

$A = 2^3 \times 3^2 \times 5$

$B = 2^2 \times 3^4 \times 7$

(a) Find the highest common factor (HCF) of A and B.

..... **[2]**

(b) Find the lowest common multiple (LCM) of A and B.
Give your answer as a product of prime factors.

..... **[2]**

(c) The integer C is such that:

- the HCF of A and C is 6
- the LCM of A and C is $2^3 \times 3^2 \times 5 \times 11$

Find the value of C.

$C =$ ..... **[3]**

10 **(a)** Let x be a rational number and y be an irrational number.

State whether $x + y$ is rational or irrational.

..... **[1]**

(b) Give an example of an irrational number y to show that y^2 can be a rational number.

[1]

(c) The nth term of a sequence is given by the expression

$$n^2 + n + 41$$

(i) Work out the 3rd term of this sequence, and state whether it is a prime number.

..... **[2]**

(ii) Explain why the 41st term of this sequence is not a prime number.

..... **[2]**

11 The number N is written as a product of its prime factors as $N = 2^3 \times 3^x \times 5^2$.

(a) The highest common factor (HCF) of N and 540 is 180.
Find the value of x.

$x =$ ..... **[3]**

(b) The lowest common multiple (LCM) of N and a number M is $2^4 \times 3^3 \times 5^2 \times 7$.
Given that $M = 2^y \times 3^z \times 7$, find the value of y and the value of z.

$y =$ ..... , $z =$ ..... **[2]**

(c) Find the smallest positive integer k such that Nk is a cube number.

$k =$ ..... **[2]**

12 **(a)** The highest common factor (HCF) of x and 12 is 4.
The lowest common multiple (LCM) of x and 12 is 120.
Find the value of x.

$x =$ ..... **[2]**

(b) **(i)** Express 1008 as a product of its prime factors.

..... **[2]**

(ii) An integer N is a perfect cube.
The highest common factor (HCF) of N and 1008 is 72.
Find the smallest possible positive value of N.

$N =$ ..... **[3]**

13 Let $m = p^3q$ and $n = pq^3$, where p and q are distinct prime numbers.

(a) Express, in terms of p and q:

(i) the highest common factor (HCF) of m and n,

..... **[1]**

(ii) the lowest common multiple (LCM) of m and n.

..... **[1]**

(b) The HCF of m and n is 15.
It is given that $m > n$.
Find the value of p and the value of q.

$p =$ ..... , $q =$ ..... **[3]**

(c) Show algebraically that the product $m \times n$ is a square number.

[2]

(d) Using your values of p and q, find the smallest positive integer k such that $m \times k$ is a cube number.

$k =$ ..... **[2]**

14 **(a)** The lowest common multiple (LCM) of two numbers p and q is 1260.
The highest common factor (HCF) of p and q is 14.
Given that $p = 70$, find the value of q.

$q =$ ..... **[3]**

(b) Consider the two numbers M and N given by:

$M = 2^x \times 3^2 \times 5$

$N = 2^3 \times 3^y \times 7$

The highest common factor (HCF) of M and N is 72.
The lowest common multiple (LCM) of M and N is 10080.
Find the value of x and the value of y.

$x =$ ..... , $y =$ ..... **[4]**

(c) Given that n is a positive integer, explain why $n^2 + n$ is always an even number.

.....

.....

..... **[2]**

15 Three numbers A, B and C are given as products of their prime factors.

$$A = 2^3 \times 3^2 \times 5 \qquad B = 2^2 \times 3^x \times 5^2 \qquad C = 2^y \times 3 \times 7$$

The highest common factor (HCF) of A and B is 180.
The lowest common multiple (LCM) of B and C is 12600.

Find the value of x and the value of y.

$x =$ ..... , $y =$ ..... **[4]**

16 Two numbers, A and B, are given as a product of their prime factors.

$$A = 2^3 \times 3^2 \times 5 \qquad B = 2^2 \times 3^3 \times 7$$

(a) Find the highest common factor (HCF) of A and B.

..... **[1]**

(b) Find the lowest common multiple (LCM) of A and B.
Give your answer as a product of its prime factors.

..... **[2]**

(c) $A \times k$ is a perfect square.
Find the smallest positive integer value of k.

$k =$ ..... **[2]**

17 Number A and Number B are given as products of their prime factors.

$A = 2^2 \times 3^3 \times 5$

$B = 2^x \times 3 \times 7$

(a) The highest common factor (HCF) of A and B is 6.
Find the value of x.

$x =$ ..... **[2]**

(b) Number $C = 2^2 \times 3^y \times 5^2$.
The lowest common multiple (LCM) of A and C is 8100.
Find the value of y.

$y =$ ..... **[3]**

(c) Here is a list of numbers.

$$\sqrt{11} \qquad -3.5 \qquad 4\pi \qquad \frac{7}{9} \qquad \sqrt{144} \qquad 0.\dot{4}$$

Write down all the irrational numbers from this list.

..... **[2]**

18 **(a)** **(i)** Write down a rational number between 0.4 and 0.5.

..... **[1]**

(ii) Write down an irrational number between 4 and 5.

..... **[1]**

(b) Give an example to show that the sum of two different prime numbers can be a square number.

[2]

(c) Write down the 6th triangle number.

..... **[1]**

(d) Find the highest common factor (HCF) of $24a^3b^2$ and $36a^2b^4$.

..... **[2]**

19 **(a)** Find the highest common factor (HCF) of 84 and 120.

..... **[2]**

(b) Find the lowest common multiple (LCM) of 84 and 120.

..... **[2]**

(c) Write down the reciprocal of 0.8 as a fraction in its simplest form.

..... **[2]**

(d) State whether $\sqrt{120}$ is rational or irrational.
Give a reason for your answer.

..... **[2]**

20 **(a)** x, y and z are positive integers.

- x is a prime number.
- y is a cube number greater than 1.
- z is a triangle number.

It is given that $x + y = z$ and $x < 10$.

Find two possible sets of values for x, y and z.

(i) First set of values.

$x =$ ..... , $y =$ ..... , $z =$ ..... **[2]**

(ii) Second set of values.

$x =$ ..... , $y =$ ..... , $z =$ ..... **[2]**

(b) p and q are two different irrational numbers.

Write down an example of p and q such that:

(i) $p + q$ is a rational number.

$p =$ ..... , $q =$ ..... **[1]**

(ii) $p \times q$ is a rational number.

$p =$ ..... , $q =$ ..... **[1]**

21 **(a)** The number 168 can be written as $2^3 \times 3 \times 7$.
The number N can be written as $2^x \times 3^2 \times 7^y$, where x and y are integers.
The highest common factor (HCF) of 168 and N is 84.
The lowest common multiple (LCM) of 168 and N is 3528.

Find the value of x and the value of y.

$x =$ ..... , $y =$ ..... **[3]**

(b) Find the exact value of N.

$N =$ ..... **[1]**

(c) Two flashing lights, A and B, are turned on at the same time.
Light A flashes every 45 seconds.
Light B flashes every 72 seconds.
They both flash together at 10:15:00.
Work out the next time they flash together.

..... **[3]**

1 **(a)** $2^3 \times 3^2 \times 7$ **[B2]**

M1 for a correct partial factorisation (e.g., 8×63 or 2×252) or a correct prime factor tree/ladder.

(b) 14 **[B2]**

M1 for identifying the needed powers 2×7 or writing $2^4 \times 3^2 \times 7^2$.

[Back to Question] *[Total Marks: 4]*

2 **(a)** $2^2 \times 3 \times 7$ **[B2]**

M1 for $2, 2, 3, 7$ correctly identified in a factor tree or division ladder

(b) 252 **[B2]**

M1 for writing $126 = 2 \times 3^2 \times 7$ or writing multiples of 84 and 126

[Back to Question] *[Total Marks: 4]*

3 **(a)** **(i)** 12 or 36 **[B1]**

(ii) 64 **[B1]**

(iii) 23 **[B1]**

(b) Prime factors: $36 = 2^2 \times 3^2$ and $64 = 2^6$ **[M1]**

LCM $= 2^6 \times 3^2 = 576$ **[A1]**

[Back to Question] *[Total Marks: 5]*

4 **(a)** 13 **[B1]**

(b) $\sqrt{7}$ **[B1]**

(c) -5 **[B1]**

(d) 0.25 **[B1]**

(e) 64 **[B1]**

[Back to Question] *[Total Marks: 5]*

5 **(a)** **(i)** 23 **[B1]**

(ii) 8 or 27 or 64 **[B1]**

(iii) 64 **[B1]**

(b) $2^2 \times 3^2$ (or $2 \times 2 \times 3 \times 3$) **[B2]**

Partial working: Correct factor tree or repeated division showing 2 and 3 as prime factors **[M1]**

[Back to Question] *[Total Marks: 5]*

6 **(a)** **(i)** 2 or 23 **[B1]**

(ii) 27 or 64 **[B1]**

(iii) 64 **[B1]**

(iv) 15 **[B1]**

(b) $2^3 \times 3^2$ **[B2]**

If 0 scored, M1 for $2 \times 2 \times 2 \times 3 \times 3$ or a correct factor tree/division **[M1]**

[Back to Question] *[Total Marks: 6]*

7 **(a)** Factor tree or repeated division showing prime factors 2, 2, 2, 3, 7 **[M1]**

$2^3 \times 3 \times 7$ **[A1]**

(b) $2^2 \times 3 \times 7$ or matching list of factors **[M1]**

84 **[A1]**

(c) $2^3 \times 3^2 \times 7$ or matching list of multiples **[M1]**

504 **[A1]**

[Back to Question] *[Total Marks: 6]*

8 **(a)** **(i)** 13 **[B1]**

(ii) 8 or -1 **[B1]**

(iii) $\sqrt{30}$ or π **[B1]**

(iv) -5 or -1 **[B1]**

(b) $\frac{5}{4}$ or $1\frac{1}{4}$ **[B2]**

M1 for $\frac{1}{0.8}$ or $\frac{10}{8}$ or $\frac{8}{10}$ seen.

[Back to Question] *[Total Marks: 6]*

9 **(a)** 36 (or $2^2 \times 3^2$) **[B2]**

Partial working: Identifies 2^2 and 3^2 as common prime factors **[M1]**

(b) $2^3 \times 3^4 \times 5 \times 7$ **[B2]**

Partial working: One incorrect power or missing prime factor **[B1]**

(c) 66 **[B3]**

Partial working: Identifies that C must contain the factor 11 from the LCM **[M1]**

Partial working: Identifies that C must contain exactly $2^1 \times 3^1$ to satisfy the HCF condition **[M1]**

[Back to Question] *[Total Marks: 7]*

10 **(a)** Irrational **[B1]**

(b) Any valid example, e.g. $\sqrt{2}$, $\sqrt{3}$, π (if $y^2 = \pi^2$ which is irrational, so this is wrong)

Only square roots of non-square rational numbers, e.g. $\sqrt{2}$ **[B1]**

(c) **(i)** 53 and Yes **[B2]**

M1 for substituting $n = 3$: $3^2 + 3 + 41$

(ii) Substituting $n = 41$ yields $41^2 + 41 + 41$ **[M1]**

This factorises to $41(41 + 1 + 1) = 41 \times 43$, which has factors other than 1 and itself, therefore it is not prime. **[A1]**

[Back to Question] *[Total Marks: 6]*

11 **(a)** 2 **[B3]**

M2 for $540 = 2^2 \times 3^3 \times 5$ and $180 = 2^2 \times 3^2 \times 5$ written as prime products

or M1 for either 540 or 180 correctly written as a product of prime factors.

(b) $y = 4$ **[B1]**

$z = 3$ **[B1]**

(c) 15 **[B2]**

M1 for identifying the needed powers to reach multiples of 3, e.g. $3^1 \times 5^1$.

[Back to Question] *[Total Marks: 7]*

12 **(a)** 40 **[B2]**

If 0 scored, allow [M1] for setting up the equation $12x = 4 \times 120$ or finding prime factors of 120 indicating $2^3 \times 5$.

(b) **(i)** $2^4 \times 3^2 \times 7$ **[B2]**

If 0 scored, allow [M1] for a correct factor tree or continuous division with at least 2 correct prime division steps.

(ii) 216 **[B3]**

If 0 scored, allow [M1] for writing $72 = 2^3 \times 3^2$.

Allow a further [M1] for deducing that N must contain $2^3 \times 3^3$ to be the smallest perfect cube with the correct HCF (power of 2 is exactly 3, power of 3 is at least 3).

[Back to Question] *[Total Marks: 7]*

13 **(a)** **(i)** pq **[B1]**

(ii) p^3q^3 **[B1]**

(b) $pq = 15$, so the prime numbers are 3 and 5 **[M1]**

$m > n \implies p^3q > pq^3 \implies p^2 > q^2 \implies p > q$ **[M1]**

$p = 5$ and $q = 3$ **[A1]**

(c) $m \times n = (p^3q)(pq^3) = p^4q^4$ **[M1]**

$= (p^2q^2)^2$ which is the square of an integer, hence a square number **[A1]**

(d) Substitutes values: $m = 5^3 \times 3^1$ **[M1]**

Recognises the power of 3 needs to be a multiple of 3, so $k = 3^2 = 9$ **[A1]**

[Back to Question] *[Total Marks: 9]*

14 **(a)** $p \times q = \text{LCM} \times \text{HCF}$ or $70 \times q = 1260 \times 14$ **[M1]**

$q = \dfrac{17640}{70}$ **[M1]**

252 **[A1]**

(b) Identifying $72 = 2^3 \times 3^2$ or $10080 = 2^5 \times 3^2 \times 5 \times 7$ **[M1]**

Comparing powers of 2 for LCM to establish $x = 5$ **[M1]**

$x = 5$ **[A1]**

$y = 2$ **[A1]**

(c) Factorising expression to $n(n + 1)$ **[M1]**

Valid statement explaining that n and $n + 1$ are consecutive integers, therefore one must be a multiple of 2 (even), making their product always even **[A1]**

[Back to Question] *[Total Marks: 9]*

15 $x = 2$ **[B2]**

$y = 3$ **[B2]**

If 0 scored, award M1 for expressing $180 = 2^2 \times 3^2 \times 5$ or $12600 = 2^3 \times 3^2 \times 5^2 \times 7$

[Back to Question] *[Total Marks: 4]*

16 **(a)** 36 (or $2^2 \times 3^2$) **[B1]**

(b) $2^3 \times 3^3 \times 5 \times 7$ **[B2]**

If 0 scored, allow [M1] for a product with at least three correct prime bases and powers, or for the correct evaluation 7560.

(c) 10 **[B2]**

If 0 scored, allow [M1] for identifying required missing powers 2×5 or $2^1 \times 5^1$.

[Back to Question] *[Total Marks: 5]*

17 **(a)** Identifies HCF as $2^1 \times 3^1$ **[M1]**

$x = 1$ **[A1]**

(b) Prime factorises $8100 = 81 \times 100 = 3^4 \times 2^2 \times 5^2$ **[M1]**

Compares highest powers of 3 between A (3^3) and C (3^y) to the LCM (3^4) **[M1]**

$y = 4$ **[A1]**

(c) $\sqrt{11}$ **[B1]**

4π **[B1]**

[Back to Question] *[Total Marks: 7]*

18 **(a)** **(i)** Any correct rational number (e.g., 0.45 or $\frac{9}{20}$) **[B1]**

(ii) Any correct irrational number (e.g., $\sqrt{17}$ or $\pi + 1$) **[B1]**

(b) Two different prime numbers identified **[M1]**

Correct sum equating to a square number (e.g., $2 + 7 = 9$ or $13 + 3 = 16$) **[A1]**

(c) 21 **[B1]**

(d) 12 or a^2b^2 seen in the final answer **[M1]**

$12a^2b^2$ **[A1]**

[Back to Question] *[Total Marks: 7]*

19 **(a)** 12 **[B2]**

If 0 scored, M1 for $84 = 2^2 \times 3 \times 7$ and $120 = 2^3 \times 3 \times 5$ OR for a common factor of 4 or 6 identified **[M1]**

(b) 840 **[B2]**

If 0 scored, M1 for $2^3 \times 3 \times 5 \times 7$ **[M1]**

(c) $\frac{5}{4}$ or $1\frac{1}{4}$ **[B2]**

If 0 scored, M1 for $\frac{1}{0.8}$ or $\frac{10}{8}$ seen **[M1]**

(d) Irrational, because 120 is not a perfect square (or $\sqrt{120}$ cannot be written as a simple fraction) **[B2]**

If 0 scored, B1 for Irrational with no/incorrect reason **[B1]**

[Back to Question] *[Total Marks: 8]*

20 **(a)** **(i)** $x = 2$, $y = 8$, $z = 10$ **[B2]**

Partial working: Lists correctly prime numbers $x < 10$ and cube numbers $y > 1$ **[M1]**

(ii) $x = 7$, $y = 8$, $z = 15$ OR $x = 2$, $y = 64$, $z = 66$ **[B2]**

Partial working: Identifies another valid combination of a prime and cube summing to a triangle number **[M1]**

(b) **(i)** e.g. $p = \sqrt{2}$, $q = -\sqrt{2}$ OR $p = 1 + \sqrt{3}$, $q = 1 - \sqrt{3}$ **[B1]**

(ii) e.g. $p = \sqrt{2}$, $q = \sqrt{8}$ OR $p = \sqrt{3}$, $q = 2\sqrt{3}$ **[B1]**

[Back to Question] *[Total Marks: 6]*

21 **(a)** $x = 2$ **[B1]**

$y = 2$ **[B1]**

M1 for $84 = 2^2 \times 3 \times 7$ OR $3528 = 2^3 \times 3^2 \times 7^2$ seen **[M1]**

(b) 1764 **[B1]**

(c) 10:21:00 **[B3]**

M2 for LCM of 45 and 72 = 360 seconds (or 6 minutes) **[M2]**

If M2 not scored, M1 for listing multiples of 45 and 72 with at least one common multiple, OR $45 = 3^2 \times 5$ and $72 = 2^3 \times 3^2$ **[M1]**

[Back to Question] *[Total Marks: 7]*